

## Read This First

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### About This Manual

This Technical Reference Manual (TRM) details the integration, the environment, the functional description, and the programming models for each peripheral and subsystem in the device.

The TRM should not be considered a substitute for the data manual, rather a companion guide that should be used alongside the device-specific data manual to understand the details to program the device. The primary purpose of the TRM is to abstract the programming details of the device from the data manual. This allows the data manual to outline the high-level features of the device without unnecessary information about register descriptions or programming models.

### Notational Conventions

This document uses the following conventions.

- Hexadecimal numbers may be shown with the suffix h or the prefix 0x. For example, the following number is 40 hexadecimal (decimal 64): 40h or 0x40.
- Registers in this document are shown in figures and described in tables.
  - Each register figure shows a rectangle divided into fields that represent the fields of the register. Each field is labeled with its bit name, its beginning and ending bit numbers above, and its read/write properties with default reset value below. A legend explains the notation used for the properties.
  - Reserved bits in a register figure can have one of multiple meanings:
    - Not implemented on the device
    - Reserved for future device expansion
    - Reserved for TI testing
    - Reserved configurations of the device that are not supported
  - Writing nondefault values to the Reserved bits could cause unexpected behavior and should be avoided.

### Glossary

[TI Glossary](#) — This glossary lists and explains terms, acronyms, and definitions.

### Related Documentation From Texas Instruments

For product information, visit the Texas Instruments website at <http://www.ti.com>.

**SPNA106** — *Initialization of Hercules™ ARM® Cortex®-R4F Microcontrollers Application Report*. Provides a brief overview and initialization procedure of the TMS570LS31x series and the RM4x series of microcontrollers in the Hercules family.

**SPNS182** — *RM46Lx30 16- and 32-BIT RISC Flash Microcontroller Data Manual*.

**SPNS183** — *RM46Lx40 16- and 32-BIT RISC Flash Microcontroller Data Manual*.

**SPNS184** — *RM46Lx50 16- and 32-BIT RISC Flash Microcontroller Data Manual*.

**SPNS185** — *RM46L852 16- and 32-BIT RISC Flash Microcontroller Data Manual*.

**SPNU551**— *Safety Manual for RM46x Hercules™ ARM® Safety Critical Microcontrollers User's Guide*. A safety manual for the Texas Instruments Hercules safety critical microcontroller product family. The product family utilizes a common safety architecture that is implemented in multiple application focused products.

**SPNU566**— *RM46x Hercules™ Development Kit (HDK) User's Guide*. Describes the board level operations of the RM46 Hercules Development Kit (HDK). The HDK is based on the Texas Instruments RM46L852 Microcontroller. The RM46 HDK is a table top card that allows engineers and software developers to evaluate certain characteristics of the RM46L852 microcontroller to determine if the microcontroller meets the designer's application requirements as well as begin early application development. Evaluators can create software to execute on board or expand the system in a variety of ways.

## Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

**TI E2E™ Online Community**— *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At [e2e.ti.com](http://e2e.ti.com), you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

**TI Embedded Processors Wiki**— *Texas Instruments Embedded Processors Wiki*. Established to help developers get started with Embedded Processors from Texas Instruments and to foster innovation and growth of general knowledge about the hardware and software surrounding these devices.

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